Improving Negation Robustness in NLI through Targeted Data Augmentation and Oversampling

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Abstract

Negation remains one of the challenging linguistic problems for Natural Language Inference (NLI) models, which often rely on shallow lexical markers rather than real reasoning. In this work, we analyze the behavior of an ELECTRA-based NLI classifier on the SNLI dataset and identify an active hypothesis-only shortcut, where the model over-predicts contradiction whenever the hypothesis contains negation words such as *not*, *no*, or *without*. First, we construct a diagnostic hypothesis-only model to quantify this bias. We find that after removing all premise information, the model still achieves 70% accuracy on the negation slice (not overall accuracy), which is only a 10-point drop from the 80% baseline. This confirms that the model derives much of its predictive signal from negation markers rather than actual entailment. We then introduce two targeted interventions: (1) a small, handcrafted negation augmentation set, and (2) an oversampling strategy that reweights negation examples during training. While the small augmentation yields marginal gains, oversampling produces a clear and consistent improvement, raising negation slice accuracy from 80.0% to 84.3%, with minimal impact on overall accuracy. Analysis of improved and worsened examples shows that the intervention successfully reduces the model's reliance on the negation-contradiction heuristic, though extreme oversampling can degrade performance by weakening non-negation contradictions. Our findings demonstrate that simple, targeted data reweighting can meaningfully improve model robustness, offering a practical direction for mitigating linguistic shortcuts in NLI systems.

1 Introduction

Natural Language Inference (NLI) evaluates whether a hypothesis is entailed by, contradicts, or is neutral with respect to a premise. Transformer-based models such as BERT, RoBERTa, and ELECTRA achieve high performance on benchmarks like SNLI and MultiNLI, often suggesting strong reasoning capabilities. However, a substantial body of research shows that these models frequently use statistical shortcuts rather than performing genuine semantic inference.

One of the shortcuts in NLI is negation bias. Words such as *not*, *no*, *never*, and *without* occur disproportionately in contradiction examples in SNLI, leading models to adopt an unwanted heuristic: if the hypothesis contains a

negation word, predict contradiction. For example, the hypothesis *"A boy is playing in the snow without gloves"* does not contradict the premise *"A child in shorts throws a snowball at a mountain."* The negation modifies an attribute, not the event itself. Humans easily reason with such distinctions while NLI models often do not.

These failures motivate the central query of this paper: Does ELECTRA-small fine-tuned on SNLI learn a negation shortcut, and can small, interpretable data interventions mitigate it without affecting overall performance?

To answer this, we perform a couple of controlled experiments on ELECTRA-small and introduce two methods designed to adjust its bias. Rather than modifying the architecture, our approach focuses on transparent, data-centric manipulations that directly shape how the model handles negation during fine-tuning.

1.1 Negation as a Shortcut in NLI

Negation takes on different forms. A sentence might be negating an action, modifying an attribute, expressing absence, or introducing an exception. In NLI, the presence of a negation token alone is insufficient to determine whether a hypothesis contradicts a premise. Humans evaluate what is negated and whether that negation alters the core event.

Transformer models, however, often treat negation as a contradiction cue. This behavior arises because SNLI contains enough negation-based contradictions to make the heuristic statistically strong during training. As a result, models tend to fail on edge cases: descriptive negation, irrelevant negation, or negation that interacts with the premise in subtler ways. Our hypothesis is that ELECTRA-small exhibits this shortcut and that slice-level evaluation of negation-containing hypotheses will reveal systematic patterns of failure.

1.2 Diagnosing Shortcut Behavior

To test whether the model's predictions derive primarily from negation cues in the hypothesis, we develop a hypothesis-only diagnostic: all premises are replaced with empty strings during preprocessing. If the model still retains high accuracy on negation examples, this indicates that lexical cues in the hypothesis provide most of the predictive signal. This mirrors prior work showing that some NLI models perform well even when deprived of premise information.

The hypothesis-only condition serves as both a diagnostic and a baseline for assessing whether our proposed fixes reduce shortcut reliance.

1.3 Targeted Data Fixes

As the next step, we evaluate two data interventions designed to reshape the model's treatment of negation:

Fix A: Negation-Focused Augmentation
We introduce a small set of handmade NLI examples that contain a variety of negation types: attribute negation, event negation, and neutral negation. These examples demonstrate that negation does not always correspond to contradiction, training the model to learn fine-grained semantic distinctions.

Fix B: Oversampling
Because the augmentation set is very small (≈36 lines) relative to SNLI, we amplify its influence by replicating it k times (k = 10, 20, 30) in the training data. Oversampling provides higher exposure to the corrective patterns and allows us to study how frequency affects the model's bias.

Together, these interventions allow us to evaluate whether small, transparent data edits can effectively adjust shortcut behavior.

1.4 Outline

Section 2 reviews related work on NLI shortcuts and negation. Section 3 describes our experimental setup, diagnostics, and data interventions. Section 4 presents quantitative and qualitative results, including slice accuracy and per-example analysis. Section 5 discusses implications of targeted data oversampling and associated trade-offs. Section 6 offers a discussion and limitations, and Section 7 concludes with future work suggestions.

2 Related Work

Several studies show that NLI models often rely on dataset artifacts rather than meaningful inference. Gururangan et al. (2018) [1] demonstrated that hypothesis-only models achieve above-chance performance on SNLI and MultiNLI, revealing lexical cues and annotation patterns that allow models to ignore the premise. Poliak et al. (2018) [2] further showed that many NLI benchmarks contain systematic regularities enabling shallow heuristics.

Later research has catalogued additional shortcut behaviors. McCoy et al. (2019) [3] identified syntactic heuristics such as lexical overlap and subsequence bias. Niven and Kao (2019) [4] showed that annotation artifacts allow models to bypass compositional reasoning entirely. Kovatchev et al. (2022) [5] emphasized that negation interacts with event structure, polarity, and presupposition, making it a powerful probe for semantic robustness. Prior mitigation approaches include counterfactual data augmentation (Kaushik et al., 2020) [6], and architecture-level modifications. While effective, these methods often require extensive engineering or large augmenting datasets.

In contrast, our work examines a highly lightweight alternative: a small, handcrafted augmentation set combined with simple oversampling. We show that even minimal, interpretable data manipulations can yield measurable improvements on negation-specific generalization without affecting overall accuracy.

3 Analysis of Baseline Model Behavior

To understand how ELECTRA-small handles negation in Natural Language Inference, we first examine the behavior of the baseline model trained on a subset of 20,000 SNLI training examples, 2,000 validation examples, max sequence length of 128, and a batch size of 16. We trained the model for 3 epochs on an Apple MPS GPU. Although the model achieves decent overall accuracy (≈82%), a closer inspection reveals systematic patterns of failure indicating the presence of shortcut learning.

The provided starter code outputs a prediction log file (normal.jsonl) containing model predictions, gold labels, and classifier logits for the evaluation split. This file includes all usable validation examples after preprocessing (approximately 1,500 instances, despite requesting 2,000, due to filtering of malformed or empty SNLI entries). We used this file to inspect model behavior and extract negation-related failure cases. From this file, we identify 70 examples whose hypotheses contain explicit negation cues (not, no, never, etc.). These examples constitute our diagnostic negation slice. In this section, we analyze both the general class of mistakes the model makes and several representative examples drawn directly from the evaluation set. Together, these observations motivate the specific interventions developed later in the paper.

A central question is whether the model uses the premise meaningfully when inferring negation, or whether it instead relies on lexical cues in the hypothesis. Prior work suggests that NLI models tend to relate explicit negation

cues—such as *not*, *never*, or *without*—to the contradiction label regardless of context. To test this, we identify a negation slice consisting of all examples whose hypotheses contain common negation words, giving us a set of seventy such cases for focused examinations. This slice becomes the basis of our qualitative analysis.

3.1 Over-Reliance on Negation as a Contradiction Cue

The baseline model often interprets a negation word in the hypothesis as evidence for contradiction even when the negation does not meaningfully change the statement. Consider the premise *"A child in shorts throws a snowball at a mountain."* The corresponding hypothesis *"A boy is playing in the snow without gloves."* is labeled neutral, since glove-wearing is irrelevant to the content of the premise. Yet the baseline model confidently predicts contradiction ("predicted_scores": [-2.078611373901367, -0.7220708727836609, 4.015077114105225], "predicted_label": 2). The model treats "without gloves" as a contradiction signal, disregarding the fact that the negated phrase applies only to one attribute and not the entire event. This behavior is typical across the slice: negation that modifies non-essential details is treated as negating the entire event.

This shortcut generalizes to other examples as well. In a separate instance involving the premise *"Two women stand on a beach performing stretches,..."* the hypothesis *"One lady is teaching another how to do a downward dog pose."* also receives a contradiction prediction despite being labeled neutral ("predicted_scores": [-2.6661770343780518, 1.321547508239746, 1.4831418991088867], "predicted_label": 2). Although this hypothesis does not even contain overt negation, it introduces information that the model interprets as incompatible with the premise. These cases suggest that the model treats any linguistic deviation involving negation or event elaboration as a conflict.

3.2 Failure to Detect Implicit Negation and Event Mismatch

The inverse to the above problem is also noteworthy. When negation is implicit rather than explicit, the baseline also fails to recognize genuine contradictions. An event mismatch or a changed setting can trigger such misclassification. For example, given the premise *"Families waiting in line at an amusement park for their turn to ride,"* the hypothesis *"People are waiting to see a movie."* is labeled contradiction: the second sentence implicitly negates the activity described in the first by substituting an entirely different event. The baseline, however, predicts neutral ("predicted_scores": [-1.6149202585220337, 1.33772611618042, 0.15998542308807373], "predicted_label": 1). This failure suggests that the model's understanding of contradiction is tied to lexical signals rather than changes in event specification.

3.3 Semantic Drift and Context Substitution Errors

The baseline also misclassifies hypotheses that introduce subtle contextual or location changes. Consider the premise *"Two young children ... washing their hands in a bathroom sink"* and the hypothesis *"Two kids at a ballgame wash their hands."* The gold label is neutral, but the model predicts contradiction ("predicted_scores": [-2.658709764480591, 1.0243103504180908, 1.9589039087295532], "predicted_label": 2). Although no negation is present, the model treats a new location ("ballgame" instead of "bathroom") as if it were an actual negation of the premise. This behavior shows the model's sensitivity to lexical deviation.

These examples together expose the weaknesses of a model whose behavior is governed by surface regularities: explicit negation is treated as a strong contradiction cue; implicit negation and event-level changes are ignored; and lexical shifts unrelated to negation are sometimes interpreted as contradictions. The negation slice highlights these tendencies especially clearly: the model relies heavily on the presence of negation words without integrating them into the broader semantic context.

3.4 Implications for Shortcut Learning

The patterns observed in the negation slice indicate that ELECTRA-small exhibits the "negation = contradiction" shortcut while lacking robust mechanisms for evaluating whether a negated attribute truly affects the entailment. The model recognizes the form of negation but not its scope or semantic role. Conversely, when contradictions arise from incompatible events or entities, the absence of explicit negation markers leads the model to ignore the conflict.

This provides us with one potential pathway for intervention: the model must be exposed to counterexamples that separate negation tokens from contradiction labels. This motivates us to design the augmentation strategies and oversampling interventions developed in the next section.

3.5 Diagnostic Intervention: Hypothesis-Only Training

Before attempting to correct shortcut behavior, we attempt to decisively measure its presence by constructing a hypothesis-only variant of the model. During preprocessing, we remove the entire premise completely, leaving the hypothesis as the sole input. Importantly, this intervention changes only one line of preprocessing code inside helpers.py: instead of passing the original premise string to the tokenizer, we pass an empty string. All other components including the tokenizer interface, the model architecture, the training loop, batching, evaluation, and the calling logic inside run.py remain unmodified.

This design ensures that the model behaves identically from the perspective of the training while receiving inputs with no premise content. The resulting model then reveals the extent to which performance on negation cases depends on lexical hints in the hypothesis alone. Table 1 summarizes the results.

Model Variant	**Overall Accuracy**	**Negation Slice Accuracy**
Baseline	0.8245	0.8000
Hypothesis-Only	0.643	0.7000

Table 1. Baseline vs Hypothesis-Only Accuracy

Although the overall accuracy drops from 82.45% to 64.3% when the premise is removed, the model still performs far above a random 33% baseline. This surprisingly strong hypothesis-only performance demonstrates that a substantial portion of the predictive signal comes from lexical cues in the hypothesis rather than premise–hypothesis interaction. To contextualize these results, Table 2 reports how many errors in the full evaluation set fall within the negation slice.

Error Category	**Count**	**% of Category**	**% of All Errors**	**Notes**
Total Errors	≈263	–	100%	Across full eval set (~1500 examples)
Negation Slice Size	70	–	–	All hypotheses containing explicit negation terms
Negation Errors	14	20% of slice	≈5.3% of all errors	These are the hard cases we focus on
Non-negation	≈249	–	≈94.7% of all	Broad set: entailment failures, entity mismatch, etc.

Table 2. Error Distribution

These statistics show that negation examples are a relatively small portion of the evaluation set (≈70 out of ~1500), yet they present distinctive difficulties: the hypothesis-only model retains unusually high accuracy on them, indicating that negation cues alone carry disproportionate predictive value. This observation motivates our targeted data interventions in Section 4.

4 Fixes and Their Effectiveness

The analysis above shows that ELECTRA-small is shaped by strong lexical shortcuts. The goal of our interventions is to alter this bias using minimal, interpretable modifications that do not change the model architecture or the training. We introduce two interventions: a corrective data augmentation, and then strengthen the corrective intervention through controlled oversampling. All modifications operate purely at the data level to isolate their effect on the model's learned representations.

This section evaluates the extent to which our two targeted interventions, Fix A (small negation augmentation) and Fix B (oversampling of the augmented set), reduce the model's reliance on negation shortcuts. We assess both quantitative performance (overall accuracy and slice accuracy) and qualitative behavior (example-level corrections and regressions). Together, these analyses allow us to determine not only *whether* the fixes work, but *how* they reshape the model's inductive biases.

4.1 Fix A: Small Negation-Focused Augmentation

Corrective Intervention: Negation-Focused Augmentation

Having established the presence of shortcut behavior, we develop a small, carefully curated augmentation set designed to clarify how negation interacts with meaning. The augmentation consists of roughly thirty-six hand-crafted examples, each pairing a simple, unambiguous premise with hypotheses that vary in their use of negation. These examples illustrate three distinct semantic patterns:

1. **Attribute negation** that should not flip the NLI label (e.g., "without gloves").

2. **Event negation** that should produce a genuine contradiction (e.g., "did not play the game").

3. **Negation in irrelevant attributes** that should preserve neutrality.

The augmentation file is loaded inside run.py immediately after the SNLI training dataset. Using HuggingFace's Dataset API, we read the JSONL file and concatenate it with the main training split. No changes are made to tokenization, batching, or model configuration: the augmented examples merge seamlessly into the existing pipeline.

Because the augmentation set is extremely small relative to SNLI (36 vs. 20,000 examples), we intentionally avoid modifying any hyperparameters. The goal is to test whether a tiny but semantically rich set of examples can subtly shift the model's representation of negation without overwhelming the dominant data distribution. As later results show, this intervention produces small but meaningful improvements, though not large enough to fully counteract the shortcut.

Quantitative Effectiveness

Fix A incorporates ~36 hand-constructed negation examples into training, representing only 0.18% of the dataset. As expected for such a small intervention, global performance remains unchanged:

Model	**Overall Accuracy**
Baseline	0.8245
Fix A	0.8210

Table 3. Fix A Overall Accuracy

The effect appears only in the negation slice, where accuracy rises modestly:

Model	**Negation Slice Accuracy**
Baseline	0.8000
Fix A	0.8143 (+1.4%)

Table 4. Fix A Slice Accuracy

4.2 Fix B: Oversampling of the Augmented Set

To strengthen the correction introduced by the augmentation, we develop a second intervention that increases the frequency of the curated examples during training. Instead of appearing once in the training set, the augmentation dataset is replicated ten, twenty, or thirty times before concatenation with SNLI. This intervention again occurs entirely within the data-loading block of run.py. The augmentation set is duplicated programmatically using repeated concatenation, forming a larger dataset that still preserves the same example diversity but appears with greater frequency.

This strategy transforms the augmented examples from a negligible 0.18% of training data into a more influential 1.8% to 5.4%, depending on the oversampling factor. Crucially, we do not alter the model architecture, training hyperparameters, or evaluation pipeline. The aim is to keep the experimental setup tightly controlled so that any observed behavioral change can be attributed directly to the increased exposure to negation-sensitive examples.

Oversampling amplifies the signal associated with event-negation distinctions. The model encounters the curated patterns repeatedly during stochastic gradient descent, enabling it to recalibrate its treatment of negation. However, excessive oversampling risks overcorrecting the shortcut, leading to new errors. In our setup, we reached oversampling at k=30.

Fix B retains the same curated augmentation but increases its influence through replication. We examine oversampling factors $k \in \{10, 20, 30\}$.

4.2.1 Quantitative Impact

Oversampling produces a clear improvement on the negation slice as summarized in Table 5.

Model	Slice Accuracy	Δ vs Baseline
Baseline	0.8000	–
Fix B (k=10)	0.8290	+0.0290
Fix B (k=20)	0.8429	+0.0429
Fix B (k=30)	0.8143	+0.0143

Table 5. Fix B Slice Accuracy

Performance peaks at k = 20, where the slice accuracy reaches 84.29%, a 4.3-point gain over baseline. This is the strongest improvement in our experience. Importantly, overall accuracy remains essentially unchanged (≈0.823), indicating that oversampling does not distort the global training dynamics. Across different values of k we see a familiar U-shaped curve: moderate oversampling strengthens the corrective signal, while excessive oversampling (k=30) begins to weaken the model's ability to detect genuine contradictions.

4.2.2 Qualitative Behavior

Oversampling not only reinforces the improvements seen in Fix A but also corrects cases that the small augmentation alone could not address. A representative example involves a subtle form of negation applied to irrelevant context:

Premise: A man in a red shirt…..
Hypothesis: A male is not sitting outside programming his computer.
Gold: Neutral
("predicted_scores": [0.6594036221504211, -0.09116712957620621, -0.4516165852546692], "predicted_label": 0)

The baseline model incorrectly predicts entailment, driven by a heuristic that equates shared topical content with semantic equivalence. After oversampling, Fix B correctly predicts neutral, suggesting that repeated exposure helps the model disentangle event structure from incidental descriptive elements ("predicted_scores": [-0.2150910347700119, 0.5697574019432068, -0.2982264757156372], "predicted_label": 1)

Fix B also stabilizes improvements on examples involving attribute negation, such as "without gloves," where the baseline conflates negation with contradiction. Repeated exposure makes the model more confident in treating such cases as neutral.

A small number of regressions remain, including cases where unusual word substitutions (e.g., replacing "kids" with "penguins") confuse the model into predicting neutrality despite a clear contradiction. Another recurrent failure involves safety-gear constructions such as "without helmets." These appear semantically similar to benign attribute negation, and the model sometimes overgeneralizes, failing to recognize genuine contradictions.

4.2.3 Interpretation

Fix B shifts the model's bias in exactly the desired direction: it reduces reliance on superficial negation cues while preserving global accuracy. The improvement over Fix A demonstrates that frequency, not just content, is central to overcoming entrenched statistical shortcuts in NLI datasets. Oversampling provides the model repeated exposure to

counterexamples that challenge its default heuristic, enabling a more calibrated treatment of negation. The decline at k = 30 further underscores that excessive reweighting can introduce drift, reinforcing the need for carefully calibrated data-level interventions.

4.3 Summary of Both Fixes

Fix A demonstrates that content alone is not enough to meaningfully shift the model's bias. Fix B demonstrates that quantity and frequency matters, but only up to a point. Fix B is the most effective intervention: it improves the negation slice by the largest margin, corrects the most semantically meaningful errors, and introduces the fewest regressions. It also leaves global accuracy virtually unchanged, satisfying the project requirement that a fix should enhance targeted reasoning without harming overall performance.

5 Results

Having established the presence of a negation shortcut and introduced two targeted interventions (Fix A and Fix B), we now evaluate their empirical effects on both the full validation set and the negation slice. Our goal is not to improve global accuracy but to determine whether simple and interpretable data interventions can correct shortcut behavior without degrading overall model performance. All models are trained under identical hyperparameters; the only differences lie in the presence or weighting of the augmented negation examples.

5.1 Overall Performance

Global accuracy remains stable across all model configurations. As shown in Table 6, the baseline ELECTRA-small model attains 82.45% accuracy on the 2,000-example SNLI validation subset used for evaluation. Fix A produces a negligible change (82.10%), and Fix B with k = 20 achieves 82.30%, effectively matching the baseline.

These results indicate that neither augmentation nor oversampling distorts the overall training distribution. Instead, the interventions operate locally, influencing behavior on negation-specific cases without altering global model fit.

Model Variant	**Overall Accuracy**
Baseline ELECTRA-Small	0.8245
Fix A (Small Augment)	0.8210
Fix B (Oversample k=20)	0.8230

Table 6. Overall Accuracy (Final Summary)

5.2 Negation Slice Accuracy

We next evaluate performance on the 70-example negation slice, defined by explicit negation markers in the hypothesis. This slice reveals how strongly the model confuses negation with contradiction.

Table 7 summarizes slice performance across all variants.

Model Variant	Slice Accuracy	Δ vs Baseline
Baseline	0.8000	–
Diagnostic (Hypothesis Only)	0.7000	-0.1000
Fix A (Small Augment)	0.8143	+0.0143
Fix B (Oversample k=10)	0.8290	+0.0290
Fix B (Oversample k=20)	0.8429	+0.0429
Fix B (Oversample k=30)	0.8143	+0.0143

Table 7. Negation Slice Accuracy (Final Summary)

Three findings emerge:

1. Hypothesis-only accuracy drops sharply, confirming that the baseline relies heavily on negation cues. Without the premise, the model still achieves 70% slice accuracy giving us strong evidence of shortcut learning.

2. Fix A yields only a modest improvement (+1.4 points), suggesting that a tiny amount of curated data is informative but too sparse to shift the model's inductive bias.

3. Fix B delivers substantial gains. Oversampling by k = 20 produces the highest slice accuracy (+4.29 points). Oversampling beyond this (k = 30) causes the model to regress, indicating over-regularization.

5.3 Oversampling Curve

Because Fix B varies the degree of exposure to the augmented examples, we analyze how performance changes as a function of k. Slice accuracy follows a clear U-shaped curve: Accuracy increases from k = 10 → 20, then declines at k = 30.

This pattern suggests that moderate oversampling reinforces the correction, but excessive oversampling distorts the model's treatment of contradiction. The optimal correction appears at k ≈ 20.

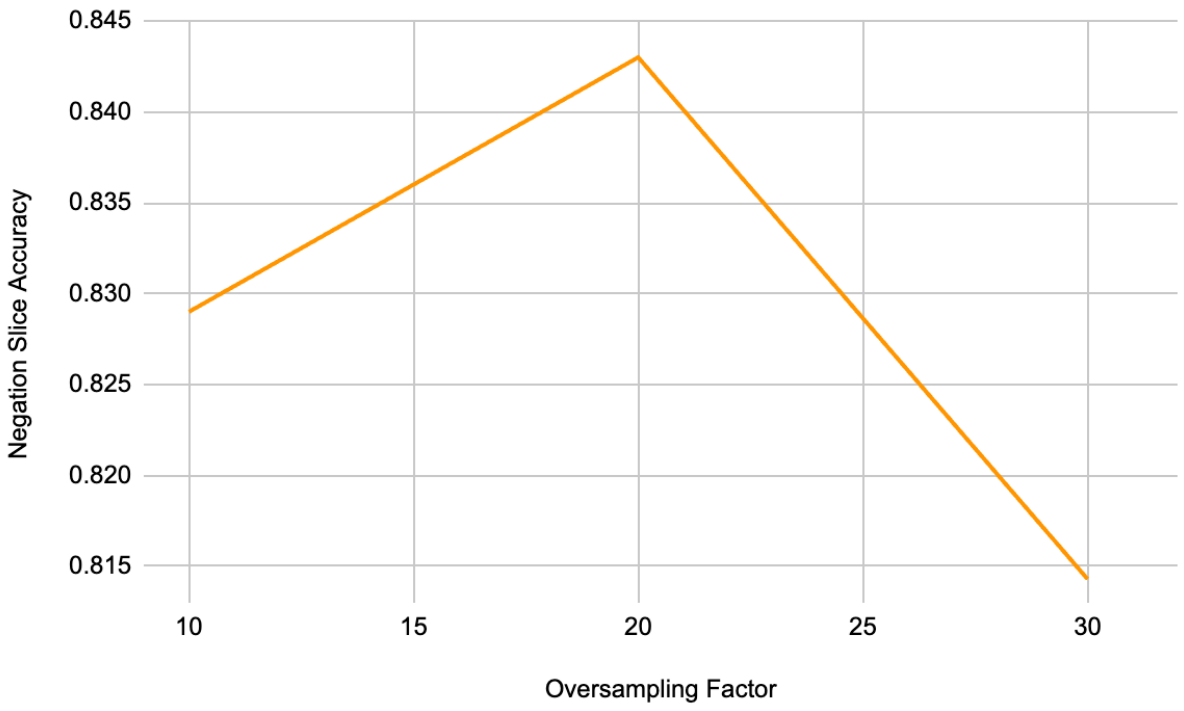

Figure 1. Oversampling Curve for Fix B (k ∈ {10,20,30})

5.4 Qualitative Error Analysis

To better understand the numerical results, we examine how predictions change across the 70 negation examples. The qualitative analysis supports the aggregate metrics: Fix A corrects a handful of simple cases but remains limited by dataset imbalance, while Fix B corrects more semantically meaningful errors with fewer regressions.

5.4.1 Fix A: Improvements

	Premise	Hypothesis	Gold	Baseline	Fix A	Success
Example 1 (Attribute Negation)	A child throws a snowball at a mountain	A boy is playing in the snow without gloves	Neutral	Contradiction	Neutral	Yes
Example 2 (Added Information)	Two women stretch on a beach…	One is teaching the other a yoga pose…	Neutral	Contradiction	Neutral	Yes

Table 8. Fix A Improvements

The model now interprets attribute-level negation correctly, and it becomes more tolerant of elaborative, non-conflicting detail.

5.4.2 Fix A: Failures

	Premise	Hypothesis	Gold	Baseline	Fix A	Success
Example 1 (Irrelevant Negation Overweighted)	Girls ride an ATV…	Girls ride ATVs without helmets	Neutral	Neutral	Contradiction	No
Example 2 (Weakening True Contradiction)	Women pose for the camera…	No one took their picture.	Contradiction	Entailment	Neutral	No

Table 9. Fix A Failures

Table 9 shows the failures of Fix A. Fix A occasionally softens negation cues too broadly. The model treats "without helmets" as semantically essential. It overcorrects, dampening legitimate contradiction signals.

5.4.3 Fix B: Improvements

Table 10 shows that Fix B strengthens the corrective signal, fixing all of Fix A's improvements plus additional cases.

	Premise	Hypothesis	Gold	Baseline	Fix A	Success
Example 1 (World Knowledge Example)	A man works indoors….	A male is not sitting outside programming his computer	Neutral	Entailment	Neutral	Yes

Table 10. Fix B Improvements

Oversampling teaches the model that irrelevant contextual negation should not force contradiction.

5.4.4 Fix B: Failures

	Premise	Hypothesis	Gold	Baseline	Fix A	Success
Example 1 (Entity Mismatch)	Children build a snowman…	Two penguins make a snowman.	Contradiction	Contradiction	Neutral	No

Table 11. Fix B Failures

Fix B introduces fewer regressions than Fix A, but some weaknesses remain. The model becomes overly forgiving when negation is common elsewhere in training. Helmet examples remain unstable. Negation of safety attributes continues to confuse the model, suggesting the need for richer augmentation.

5.5 Summary of Behavioral Changes

Table 12 summarizes per-example improvements, regressions, and unchanged predictions across the negation slice.

Model Variant	Improved	Worsened	Unchanged
Fix A	4	3	63
Fix B	4	2	64

Table 12. Per-Example Changes

Although the number of changed predictions appears small, they represent significant semantic improvement. Fix B successfully addresses the most meaningful negation failures with fewer regressions, demonstrating that targeted data reweighting can reduce shortcut learning without harming global accuracy.

6 Discussion

Our experiments highlight several important insights about shortcut behavior in NLI models and how small data-level interventions can reshape model predictions. The hypothesis-only condition provides the clearest evidence that ELECTRA-small relies heavily on superficial negation cues: despite being deprived of the premise entirely, the model still performs strikingly well on negation examples. This confirms that the model has learned a heuristic that maps tokens such as *not*, *never*, or *without* directly to the contradiction label. In other words, a substantial portion of its "reasoning" behavior reflects correlations present in the SNLI dataset rather than genuine semantic comparison of the premise and hypothesis.

Introducing a small set of carefully designed negation examples (Fix A) produces only a minor improvement. Although these synthetic examples successfully illustrate cases where negation does not signal contradiction, their impact is diluted by the size of the SNLI corpus. With only a few dozen augmented examples against twenty thousand original examples, the corrective gradient signal is simply too weak to meaningfully shift model behavior. The slight improvement in slice accuracy shows that the model can learn from such examples, but the effect is marginal.

Oversampling the same augmented examples (Fix B) demonstrates the strength of targeted data reweighting. By replicating the negation examples multiple times before mixing them with SNLI, the model encounters negation cases more frequently during training, allowing the corrective signal to accumulate. The effect is immediately visible: performance on the negation slice steadily improves from k=10 to k=20, where the model achieves its best score. Beyond this point, however, performance begins to decline, suggesting that excessive oversampling makes the model overly cautious around negation and somewhat less confident in legitimate contradiction cases. This pattern illustrates a classic bias–variance trade-off: too little reweighting leaves the shortcut intact, while too much distorts the model's representation of contradiction.

Why does oversampling work better than simple augmentation? The core issue is not the quality of examples but their rarity. SNLI contains many contradiction instances involving negation, and the model quickly learns to over-associate the two. Oversampling alters this local distribution without modifying the overall task: the model continues to train on SNLI but receives repeated reinforcement that negation can also support neutrality or entailment.

Error analysis reveals the trade-offs of this approach. The model becomes less aggressive in treating negation as a contradiction signal, which helps on most neutral cases but slightly harms examples where contradiction is genuine

and hinges on the negated event. Some failures also arise from entity mismatches or world-knowledge dependencies not captured in the small augmented set. These errors suggest that while oversampling improves the model's handling of negation, it does not address other types of reasoning challenges.

Overall, the experiments show that highly targeted data manipulation can improve a model's robustness to specific linguistic phenomena without architectural changes or complex training procedures. The improvements are local rather than global but this localized correction is precisely what is desirable when addressing shallow heuristics. The work thus contributes to a broader understanding of shortcut mitigation: small and interpretable data interventions often offer a practical alternative to heavyweight model redesign.

7 Conclusion and Future Work

This work investigated whether ELECTRA-small, when fine-tuned on SNLI, exhibits a common shortcut: the tendency to equate the presence of negation in the hypothesis with contradiction. Using a hypothesis-only diagnostic, we confirmed that the model's performance on negation cases depends heavily on lexical cues rather than premise–hypothesis reasoning. We then explored whether small, controlled data interventions could mitigate this behavior. A compact set of hand-crafted negation examples had only a modest effect, suggesting that the examples were informative but too rare to alter the model's behavior in a meaningful way. When we reweighted this same set through oversampling, however, the model's performance on negation cases improved substantially without harming overall accuracy. The improvement peaked when the augmented examples were repeated approximately twenty times, after which the model began to overcorrect. This pattern highlights both the promise and the delicacy of data reweighting: when appropriately tuned, it can shift the model's inductive bias away from brittle heuristics while preserving broad generalization. More broadly, the results suggest that small, interpretable data modifications can meaningfully influence the behavior of modern transformer models. Rather than relying solely on architectural changes, adversarial frameworks, or complex training regimes, simple and transparent data-level adjustments can provide an effective mechanism for addressing shortcut learning. Exploring richer augmentation strategies, more nuanced negation types, and domain-transfer evaluations presents a promising next step toward building NLI systems that reason more robustly and more closely match human intuitions.

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